

FinPulse:Real Time Financial Data Intelligence Platform

Report on Data Model

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1 INTRODUCTION

FinPulse addresses a fundamental problem in personal finance: transaction data is fragmented across heterogeneous sources – bank SMS alerts, UPI CSV exports (Google Pay, PhonePe), credit card PDF statements, and manual wallet entries – each with a distinct format, schema, and retrieval path. No existing consumer solution unifies these into a real-time, ML-augmented intelligence layer. This report gives idea about the main dimensions of data like – data sources, data types, retrieval mechanisms, storage architecture, and data arrangement.

2 DATA SOURCES, TYPES & RETRIEVAL

Every source is ingested via a dedicated Kafka producer and published to a separate topic, decoupling ingestion from all downstream components. This topology means that a PDF parser failure does not affect SMS or UPI ingestion – a critical reliability requirement given that each source has a different failure mode and rate.

Table 1: Data sources, formats, retrieval mechanisms, and Kafka topic routing

Source	Format	Retrieval	Data Type	Kafka Topic
Bank SMS	Unstructured text	Twilio web-hook + regex	String → parsed fields	raw.bank.sms
UPI Apps	CSV export	File upload / GPay API	Tabular, semi-structured	raw.upi.csv
Credit Card	PDF statement	pdfplumber extraction	Semi-structured	raw.card.pdf
Manual / Wallet	REST JSON body	User API entry	Structured JSON	raw.manual

3 DATA STORAGE ARCHITECTURE

FinPulse adopts a Medallion (Bronze → Silver → Gold) lakehouse architecture [3].

3.1 Bronze Layer – Delta Lake (Raw)

All Kafka data is stored in Delta Lake, partitioned by source and date (year/month/day). At this stage, data is kept in its original form as STRING or raw JSON, so parsing errors or schema changes do not cause data loss. The raw data can always be reprocessed if needed. Delta Lake also provides ACID transactions and time-travel support [3]. This partitioning makes it faster to read data for a specific source and day. Delta Lake was chosen over Iceberg and Hudi because it fits FinPulse’s mix of frequent small writes and large batch reads [4].

3.2 Silver Layer – dbt-transformed Parquet (Delta)

The silver layer contains two dbt models: stg_transactions for type casting, renaming, and removing nulls, and int_categorized for category prediction, merchant normalization, and deduplication. The data is stored in Delta Lake in a clean, typed format. Each transaction gets a SHA-256 key based on user, source, amount, and timestamp, which helps prevent duplicate records when the pipeline is retried. (SHA-256 is a hashing algorithm and it converts some data into a fixed-length string of characters called a hash.)

3.3 Gold Layer – BigQuery Analytical Warehouse

The gold layer uses BigQuery with a star schema for analytics. Its columnar storage makes month- and category-based GROUP BY queries 40–100× faster than row-based RDBMS systems [4]. The warehouse is partitioned by transaction_date and clustered by user_id and category_key, reducing dashboard query costs by up to 90% for per-user time-series queries. dbt incremental models load only new records, avoiding full-table refreshes.

3.4 Streaming State – Spark Checkpoints

Spark Structured Streaming maintains stateful per-user, per-category windowed aggregations in checkpoint storage on GCS/HDFS. This provides exactly-once semantics via offset tracking and idempotent sink writes.

4 END-TO-END PIPELINE ARCHITECTURE



Figure 1: FinPulse end-to-end data pipeline

5 DATA ARRANGEMENT

The warehouse data arrangement follows a Kimball-style star schema [7]. This is the canonical arrangement for FinPulse’s dominant query pattern: dimensional aggregations (spend by category over time, merchant-level velocity, per-user anomaly history). A star schema delivers these through a single fact table joined to small, pre-aggregated dimension tables – queries that require only one or two joins, not the multi-hop graph traversals needed for normalized 3NF designs.

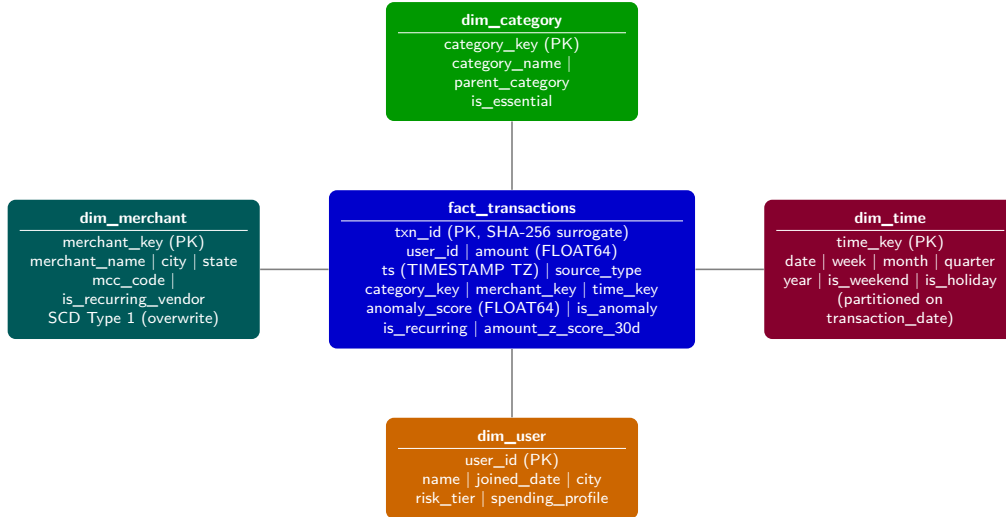


Figure 2: Star schema – fact_transactions (centre) joined to four dimension tables. Each FK is a SHA-256 surrogate key.

5.1 Fact Table – fact_transactions

One row per transaction. Key measures: amount (FLOAT64 – never INT, since UPI transactions include paise fractions), ts (TIMESTAMP WITH TIME ZONE, UTC-normalized to handle IST/UTC offsets), anomaly_score (FLOAT64 $\in [0, 1]$), is_anomaly (BOOLEAN), is_recurring (BOOLEAN), amount_z_score_30d (FLOAT64). Foreign keys reference all four dimensions.

5.2 Dimension Tables

- **dim_category:** Spending category hierarchy following RBI MCC taxonomy (18 top-level categories). SCD Type 1 (overwrite) – current categorization accuracy matters more than history.
- **dim_merchant:** Merchant name (normalized lowercase), city, state, MCC code, is_recurring_vendor flag. SCD Type 1. Merchant normalization at int_categorized layer prevents the “Swiggy” / “SWIGGY” / “Swiggy India” cardinality explosion common in UPI data.
- **dim_time:** Full calendar dimension (date, week, month, quarter, year, is_weekend, is_holiday). Supports all time-series aggregations without any date arithmetic in the query layer.
- **dim_user:** user_id, name, joined_date, city, risk_tier, spending_profile. risk_tier (LOW/MEDIUM/HIGH) is updated by the ML pipeline to segment anomaly thresholds per user – high-variance spenders get wider normal bands.

6 ETL / DBT TRANSFORMATION LAYERS

Three dbt model layers enforce strict separation of concerns. Staging models (stg_transactions) perform only type casting, column renaming, null filtering, and source_type tagging – zero business logic. This means any downstream bug is traceable to a single upstream model. Intermediate models (int_categorized) apply the ML category classifier, normalize merchant names, generate SHA-256 surrogate keys, and deduplicate on (user_id, source_type, amount, ts). Fact and dimension models (fct_transactions, dim_*) apply SCD logic, final schema enforcement, and partition-compatible inserts to BigQuery. dbt tests (not_null, unique, accepted_values) run at each layer boundary.

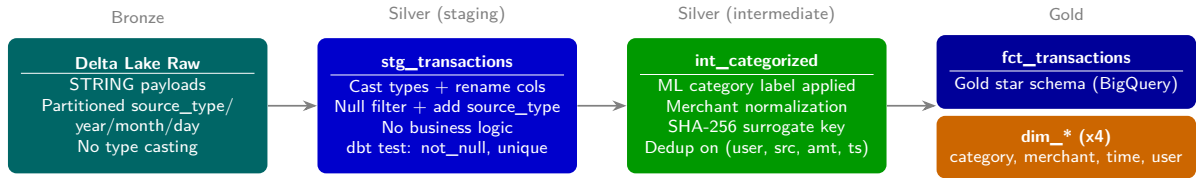


Figure 3: dbt medallion model layers: Bronze (Delta raw) → Silver staging → Silver intermediate → Gold (fact + dims)

7 ML ANOMALY DETECTION DATA MODEL

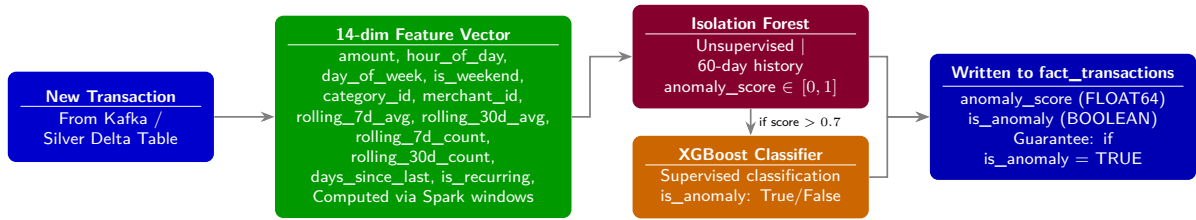


Figure 4: ML anomaly scoring pipeline – 14-dim feature vector → Isolation Forest → XGBoost → written to fact_transactions

The ML layer consumes from the silver-layer Delta table via Spark Structured Streaming. Each incoming transaction is featurized into a 14-dimensional vector computed from Spark 5-minute tumbling windows: [amount, hour_of_day, day_of_week, is_weekend, category_id, merchant_id, rolling_7d_avg_amount, rolling_30d_avg_amount, rolling_7d_txn_count, rolling_30d_txn_count, amount_z_score_7d, amount_z_score_30d, days_since_last_txn_in_category, is_recurring_merchant].

An Isolation Forest model (unsupervised, trained on 60-day rolling history) produces anomaly_score ∈ [0, 1]. If score > 0.7, an XGBoost classifier evaluates the transaction to set the binary is_anomaly flag. Both values are written back to fact_transactions. This hybrid approach combines the high recall of Isolation Forest with the targeted precision of XGBoost, minimizing false positive alerts on financial dashboards.

8 CHOSEN SOLUTIONS

Table 2: Chosen solutions and reasons

Chosen Solution	Why
Delta Lake + BigQuery	Keeps raw data safe and supports fast analytics.
Kafka + Spark Streaming	Supports real-time data processing and fraud detection.
BigQuery with partitioning and clustering	Makes large analytical queries faster and reduces query cost.
Delta Lake + BigQuery	Provides reliable storage and efficient joins for analytics.
Spark Structured Streaming	Handles both batch and streaming work in one system.

9 CONCLUSION

The FinPulse data model is a lakehouse architecture: Delta Lake bronze for durable raw preservation, dbt-modelled silver for typed and deduplicated normalized data, BigQuery gold for analytical star-schema querying, and Spark Structured Streaming for stateful real-time anomaly scoring. Four production-grade diagrams document the full pipeline, the schema, the dbt transformation chain, and the ML scoring model.

REFERENCES

Updated Tracker is given here: [CLICK HERE](#).

NOTE : Diagrams are made with Latex Code

- [1] Intuit Inc. / FinanceApps.guide, “Mint Shutting Down March 23, 2024,” Mar. 2024.
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